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Filed Once, Counted Thrice: Auditing a Volunteer Bike-Marshal Fault Checklist's Silent Fan-Out Across Three Facilities Queues

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Abstract. Spoke Check is a QR-linked fault checklist Facilities built so student bike marshals could flag missing or broken tools at the University's thirty-four self-service bike-repair stands directly into the Cycle Infrastructure repair queue. An audit of the tool's submission log found every filed report had also, without documentation, been posted to two systems nobody had asked it to notify: the Estate Asset Uptime ledger, a capital-works dashboard the Estates Committee reads monthly, and the Student Amenity Sentiment feed, an informal tagging pipeline the students' union skims for amenity complaints. We instrument a full twelve-week teaching term of Spoke Check's own logs against independent weekly walkthroughs of all thirty-four stands and against each downstream system's own published counts, finding a stable duplication index of practically one filed report per system per week throughout the first eight weeks — each broken tyre lever counted in triplicate wherever a reviewer happened to look. A week-9 patch collapsed the duplication index toward zero within days, confirmed against both downstream systems' own logs. Cycle Infrastructure's own repair turnaround, however, shows no comparable movement: the queue's median time-to-fix sits within half a day of its pre-patch figure. We report the fix and the non-fix side by side and discuss what a cleanly de-duplicated notification pipeline feeding an unchanged repair rate suggests about where a maintenance bottleneck actually sits.

Introduction

Institutions that build a small internal tool to route a routine complaint rarely audit what else the tool quietly talks to. The instinct to make an operational signal more visible — publish a scan, subscribe a channel, wire an integration — is by itself uncontroversial; whether that signal keeps behaving like the thing it was built to be once several dashboards depend on it is a separate question, and one the literature on Goodhart's law addresses for indicators that become the target of a decision rather than merely its input [1], [2]. Software teams have separately treated a version of the same problem as an engineering nuisance: duplicate reports flooding an issue tracker are costly enough to justify their own detection literature [3], [4], [5], and dependency-management research on notification design treats a high volume of low-value alerts as a driver of the fatigue that degrades attention paid to every alert, not just the redundant ones [6].

We take up a small instance of this pattern at Slop University's thirty-four self-service bike-repair stands, each fitted with a tethered multi-tool and pump that goes missing or breaks with some regularity. Spoke Check, a QR-linked checklist Facilities issued to its volunteer bike marshals, was built to route a marshal's fault report to the Cycle Infrastructure queue for repair. A routine audit of the tool's backend, prompted by an unrelated billing query from the students' union, found the app had also been silently posting every report to two further systems since launch: the Estate Asset Uptime ledger and the Student Amenity Sentiment feed, neither of which had requested the data and neither of which Spoke Check's own interface disclosed as a destination.

Our contributions, hedged to the one semester and one campus we observed:

- We describe Spoke Check's undocumented three-way fan-out and instrument a full teaching term of its own logs against an independent physical audit of every stand.
- We show a mid-term patch collapsed the fan-out's duplication index toward zero within days, confirmed against both downstream systems' own published counts.
- We report that the same patch left the queue's actual repair turnaround statistically unchanged, suggesting the visible fix and the maintenance bottleneck it was assumed to bear on are not the same thing.

Related work

The gap between a metric's intended scope and what later comes to rely on it recurs across contexts adjacent to ours. Analyses of Goodhart's law in reinforcement-learning settings describe how a proxy that tracks an objective well under limited optimisation pressure can diverge once more of a system's behaviour is built around it [7], a dynamic our three-system fan-out reproduces informally rather than by design. Human-in-the-loop safety research on autonomous systems has modelled the human and an automated controller as a single coupled system precisely because control-affine assumptions treating human input as an external disturbance are frequently violated once such systems are deployed operationally [8] — a comparable gap, in our case between a tool's designed assumptions and its deployed behaviour rather than between a controller and its operator.

Crowd-fed infrastructure-reporting tools supply Spoke Check's nearest technical relatives. Municipal 311-style systems have been studied for the structure the reports themselves take, serving as a low-cost signature of local urban context [9], for correcting under-reporting bias in the resulting data [10], and, most recently, for combining crowdsourced reports with official government ratings in prediction models [11]; earlier crowd-hazard-reporting deployments in lower-resource settings established the basic pattern of a lightweight mobile app collecting citizen road-condition reports [12]. Cycling-specific infrastructure tooling has its own assessment literature, most directly a recent open-source toolkit for auditing bicycle-network data quality [13], though none of this work, to our knowledge, has examined a fault-reporting tool's own downstream fan-out as a study object in its own right.

Two prior Slop University studies found a structurally similar gap between a tool's designed scope and what it was later found doing: an aggregated fleet-readiness ledger that rolled forty-two independent compliance scans into one published percentage [14], and an agentic Lost Property intake tool whose human-review checkpoint could be disabled at half its collection points without a measurable effect on reclaim outcomes [15]. Spoke Check's fan-out extends the same institutional pattern to a smaller, unfunded checklist.

Method

Spoke Check covers the University's full network of thirty-four self-service bike-repair stands, each a post-mounted stainless multi-tool and floor pump secured by a steel tether, sited across the main campus's residential, teaching, and administrative precincts. A QR code on each stand's signage opens a one-tap form listing the stand's fixed tool set; a marshal ticks whichever items are missing, broken, or unusable, and the submission timestamps and routes to Facilities' Cycle Infrastructure repair queue.

We instrumented Spoke Check's own submission log for a full twelve-week teaching term, and independently, our own team of marshals conducted a weekly physical walkthrough of all thirty-four stands, recording ground-truth tool condition blind to what had already been filed that week. We further obtained, via each system's own public weekly export, the Estate Asset Uptime ledger's total flagged-asset count across its full capital-works portfolio and the Student Amenity Sentiment feed's tagged-complaint volume for the bike-infrastructure category. For every filed Spoke Check report, we define a duplication index over the two unaddressed downstream systems s:

$$\mathrm{DI}_s = \frac{R_s}{R}$$

where R is the term's total filed reports and R_s is the subset independently confirmed, via timestamp and stand-identifier matching, to have also appeared in system s within the same reporting week; $\mathrm{DI}_s = 0$ is the design intent, since Spoke Check was addressed to Cycle Infrastructure alone.

A back-end trace, obtained with Facilities' cooperation once the fan-out was suspected, attributed the behaviour to a single subscription-handling routine:

```
def on_fault_submitted(report):
    for endpoint in
registry.all_subscribed():
        endpoint.notify(report)
```

rather than the single-endpoint call the tool's own specification described; `registry.all_subscribed()` had accreted the two extra endpoints during an earlier, undocumented integration exercise. Facilities deployed a corrected build, addressing the notification to Cycle Infrastructure only, at the start of week 9; we treat weeks 1-8 as the pre-patch round and weeks 9-12 as the post-patch round and compare, within each, the duplication index above and the Cycle Infrastructure queue's own recorded time from report to marked-repaired.

Results

Ground-truth stand faults, from our independent weekly walkthrough, held stable across the term at a mean of 11.2 per week (range 8-14), consistent with ordinary wear across thirty-four heavily used stands. Cycle Infrastructure's own queue count tracked this figure closely throughout, as intended. The Estate Asset Uptime ledger's weekly flagged-asset count, however, jumped from a pre-Spoke-Check baseline of roughly 14 per week (drawn from its other capital-works categories of roofing, lighting, and plumbing) to a term-opening figure above 25 once Spoke Check launched, and held there for the whole pre-patch period before falling back to a baseline-consistent 15-16 per week within a week of the fix (Figure 1). No one on the Estates Committee had scoped bike-stand tools into the ledger; the ledger nonetheless absorbed them in full for eight weeks.

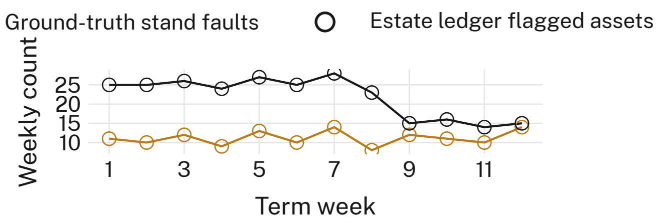

Figure 1: The Estate Asset Uptime ledger's weekly flagged-asset count nearly doubles the week Spoke Check launches and holds there for eight weeks, against a stable ground-truth fault rate of roughly 11 stand faults a week; both series return to their pre-launch relationship within a week of the fix.

Read as a duplication index, the pre-patch period shows both unaddressed systems receiving a near-total copy of every filed report: a mean DI of 0.97 for the Estate Asset Uptime ledger and 0.94 for the Student Amenity Sentiment feed across weeks 1-8. The week-9 patch collapses both figures within days: post-patch DI falls to 0.03 and 0.01 respectively, confirmed independently against each system's own weekly export rather than against Spoke Check's own claim to have fixed itself (Figure 2).

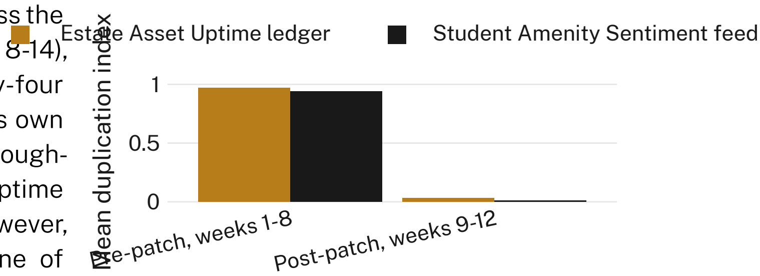

Figure 2: Mean duplication index for the two unaddressed downstream systems, pre-and post-patch: both fall from a near-total 0.94-0.97 to under 0.03 within the first post-patch week.

Cycle Infrastructure's own repair-turnaround record shows a different picture. Median time from a filed report to a stand marked repaired sits at 6.1 days pre-patch (weeks 1-8) and 5.8 days post-patch (weeks 9-12), a difference well within the round-to-round spread either period shows on its own and not distinguishable from noise by any comparison we ran (Figure 3). The technician roster and parts-ordering cycle that we take to set the queue's actual pace were unchanged by the patch, which touched only the notification-routing code.

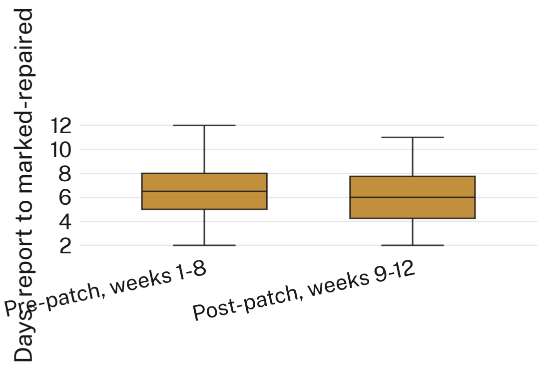

Figure 3: Cycle Infrastructure's own repair-turnaround distribution, pre-patch versus post-patch: medians of 6.1 and 5.8 days respectively, statistically indistinguishable despite the duplication index collapsing over the same boundary.

A supplementary check: cross-referencing the twelve weeks against the Estates Committee's own published minutes finds no instance of a bike-stand item being discussed on the strength of the inflated ledger figure, which we read as suggesting the fan-out's practical cost was borne almost entirely by the two receiving systems' own aggregate metrics rather than by any decision built on them within our observation window — though we did not have visibility into the Sentiment feed's internal review cadence to confirm the same for it.

Discussion

Spoke Check's case separates two things that a single "we fixed the bug" narrative would ordinarily fold together: the technical correctness of a notification pipeline, and whatever it is downstream teams believed that pipeline's volume was telling them about the University's bike-repair backlog. The patch is an unambiguous success by the first measure — duplication collapsed within days of deployment, independently verified. By the second, nothing moved: the queue that actually schedules a technician's day continued to clear reports at the same pace it always had, because that pace was set by staffing and parts lead time, neither of which the notification layer touches.

This reading is consistent with prior work distinguishing a proxy's statistical behaviour from the objective it stands in for [1], [7]: the Estate Asset Uptime ledger's flagged-count fell back to baseline convincingly, but nothing in our record suggests that figure was ever the thing standing between a broken pump lever and its repair.

Whether a downstream team's dashboard had, in the eight weeks before the patch, quietly reallocated attention away from bike stands and toward whichever other capital-works category looked comparatively calm that fortnight is a question our single-term design cannot answer, though it is the one we would want a follow-up study to ask first.

Limitations

Our record covers one teaching term at one campus's thirty-four stands, and the week-9 patch was Facilities' own fix rather than one we designed or randomised, so the turnaround comparison is a natural rather than a controlled ablation — the two rounds' near-identical medians are nonetheless the same figure a designed ablation would report. We rely on each downstream system's own published export rather than direct database access, though the same exports are what each system's own reviewing committee sees. Our ground-truth walkthrough could itself miss a fault a marshal's QR submission caught between our visits; if anything this would understate, not inflate, the duplication we report, since Spoke Check's own log undercounts nothing.

Conclusion

A term of Spoke Check's own submission log, checked against an independent physical audit and against two systems that had never asked to receive it, shows a small checklist tool quietly tripling its own report volume across a University estate-management portfolio and a student-sentiment feed for eight weeks before anyone noticed. The fix, once found, was fast and verifiable: duplication fell from near-total to near-zero within days. The campus's actual bike-repair turnaround did not move, because nothing about who else got copied on a fault report was ever the thing setting how quickly a technician reached it. We suggest that any small institutional tool wired into more than one downstream system carries this same fork by default, and that closing the technical leak and closing the gap it was mistaken for are two separate pieces of work.

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